MILITARY SPECIFICATION
RESOLVER, ELECTRICAL, AC
SPECIAL MOUNTING FOR USE IN
POLARIS FLEET BALLISTIC MISSILE WEAPON SYSTEM
TYPE 28RU42N8

1. SCOPE

1.1 This specification establishes the requirements for Resolver, AC, Type 28RU42N8, Special Mounting, hereinafter termed resolver.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-4456	Shock, Variable, Duration, Method and Apparatus for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment,
MIL-R-21530	Resolver, Electrical, AC, General Specification for

DRAWINGS

Bureau of Naval Weapons

2104021	Resolver, Gyro Error

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. - The resolver specified herein shall be in accordance with MIL-R-21530 with the exception of the following paragraphs:

3.4.23, 4.7.4.3.6, 4.8.18.3	Friction torque
3.4.24, 4.7.4.3.5	Brush slip ring assembly
3.5.1	Vibration, operating condition
3.5.3	Shock, operating condition
3.5.5, 4.8.6	Tropical fungus environment
3.5.6, 4.8.5	Salt water (corrosion)

3.1.1 Insulation resistance. - Paragraph 3.4.9 of MIL-R-21530 shall apply with the exception that the resistance value specified in 3.3 herein shall apply.

3.2 The resolver shall be of the design, construction, and physical dimensions as shown on Dwg. 2104021 and as specified herein. The design and construction of the resolver shall not be changed after acceptance and approval of the preproduction sample or of the first production lot without prior approval of the procuring activity.

3.3 Specific characteristic requirements. - The resolver shall have specific characteristic requirements as follows when measured with the inner and outer members eccentric, skewed and axially displaced as specified on Dwg. 2104021.

Primary winding	Stator
Rated primary voltage both windings	25 v maximum
Test voltage primary	25 v ± 2 percent
Test frequency	800 cycles per second (cps) ± 0.5 percent
Nominal input frequency	800 cps
Temperature rise	20 degrees centigrade (C) max at 25 degrees C ambient
Weight	172 ± 15 grams
Primary current at 25 volts	7 milliamps nominal
Primary power at 25 volts (each winding)	0.06 watts nominal
Stator impedance, Zso	1280 ± 40 percent + j 3600 ± 10 percent

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Rotor impedance, Zrs	850 ohms max. complex impedance
Impedance balance	300 Z-sec. max (resistive loads only)
Transformation ratio (Ke)	1.000 ± 5 percent
Phase shift, stator to rotor	4.0 ± 0.5 degrees
Phase shift vs rotation	± 1.5 degrees, max.
Phase shift vs voltage	± 0.5 degrees, max.
Primary tuned open circuit impedance	11000 ohms minimum
Total null voltage	100 millivolts root mean square (v rms) maximum at 22 ± 2 degrees C
Fundamental frequency null voltage	80 millivolts rms maximum at 22 ± 2 degrees C
Angular error	60 minutes of arc maximum at 22 ± 2 degrees C
Null shift (Para. 3.5.8 of MIL-R-21530)	60 minutes of arc maximum
Angular intervals for measurement of accuracy	15 degrees
Angular error vs voltage	30 sec. max.
Angular error vs vibration	150 sec. max.
Angular error vs humidity	75 sec. max.
Insulation resistance (minimum)	
(a) Windings to case	100 megohms
(b) Between windings	10 megohms

3.4 Environmental requirements

3.4.1 Temperature, operating. - Ambient temperature between the limits of 57 ± 5 degrees C.

3.4.2 Vibration, operating. - The resolver shall be capable of withstanding the vibrational forces specified in table I in each of the three mutually perpendicular planes at 22 ± 3 degrees C, in accordance with paragraph 4.8.2 of MIL-R-21530 and the angular accuracy shall not exceed the tolerances given in 3.3 herein.

3.4.3 Shock, operating. - The resolver shall be capable of withstanding each of the following half sine wave shock pulses in three mutually perpendicular planes at 22 ± 3 degrees C in accordance with paragraph 4.8.9 of MIL-R-21530. Test method and apparatus shall be in accordance with MIL-S-4456.

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Table I. Vibration During Operation

Frequency (cps)	Acceleration amplitude	Frequency sweep rate	Test Duration
5 to 15	0.030 ± 0.006 inch peak to peak	Per MIL-STD-167, type I	Per MIL-STD-167, type I
16 to 25	0.020 ± 0.004 inch peak to peak	Per MIL-STD-167, type I	Per MIL-STD-167, type I
30 to 2000 sinusoidal	5 grams ± 3 db	Logarithmic; from 30 to 2000 in two minutes	Two minutes
100 to 200 sinusoidal	10 grams ± 3 db	Logarithmic; from 100 to 200 in 20 seconds	20 seconds

3.4.3.1 Two impact shocks in each plane of 35 g ± 10 percent peak amplitude and 11 milliseconds ± 10 percent duration.

3.4.3.2 Two impact shocks in each plane of 50 g ± 10 percent peak amplitude and 3 milliseconds ± 10 percent duration.

3.4.4 Acceleration, operating. - The resolver shall be capable of withstanding acceleration in each of three mutually perpendicular directions consisting of 20 g maximum when tested at 22 ± 3 degrees C in accordance with Procedure I of MIL-E-5272. The resolver shall show no damage after acceleration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. - Sampling and inspection shall be in accordance with MIL-R-21530.

4.2 Angular error test. - Preproduction and periodic production samples shall require an additional angular accuracy test with S_1, S_2, R_1, and R_2 terminals grounded. When tested in this manner, the angular error shall not exceed the value

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specified in 3.3 herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-21530.

6. NOTES

6.1 The notes of MIL-R-21530 are applicable herein.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-21530/18(WP) RESOLVER, ELECTRICAL AC TYPE 28RU42N8

ORGANIZATION *(Of submitter)*	CITY AND STATE -

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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